

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			(Kelp needs light for) photosynthesis	1			1		
	(b)			Can only {harvest/ take/ cut} up to 0.5 m/day/ Cannot {harvest/ take/ cut} more than 0.5 m/day/ cannot harvest more than they grow in a day			1	1		
	(c)	(i)		Both scales correct + axes labels (rate of flow of oxygen + mass of iodine + units (dm ³ / minute) + (mg/kg) (1) All plots correct = 2 marks ± <1 small square 5/6 plots correct = 1 mark 0/1/2/3/4 plots correct = 0 marks Line joining the plots drawn with a ruler (1)		1 2 1		4	4	
		(ii)		no effect/ stays the same/ stays at 780 (mg/kg)			1	1		
		(iii)		Results are reported as mg/kg			1	1		1
		(iv)		$1.229091 \times 10^4 = 2$ marks Accept any correct rounding If incorrect award 1 mark for 1.2290×10^4 (incorrect rounding) 676/ 0.055 (method) 12 290.91 (not standard form) 12 291 (rounded not standard form) 12.29091×10^3 (any incorrectly expressed standard form)		2		2	1	
		(v)		Active {transport/ uptake} (1) Reference to respiration (1) Reference to {ATP/ energy} used (1) reject {make/produce} energy	3			3		

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	(d)			Any one (×1) from - time period of the experiment (OWTTE) (1) - salinity/ concentration of the water (1) - light (1) - pH (1) - carbon dioxide concentration (1) - nutrient content (1) - volume of water (1) - size of flask (1)			1	1		1
				Question 4 total	4	6	4	14	5	2